

Nursing Mothers:

The ingredients in the combination may enter breast milk and is best avoided in nursing mothers.

UNDESIRABLE EFFECTS:

Hypoglaemia: As a result of the blood-sugar lowering action of glimepiride, hypoglaemia may occur and may also be prolonged.

Eyes:

Especially at the start of treatment, temporary visual impairment may occur due to the change in blood sugar levels.

Digestive tract: occasionally nausea, vomiting, sensations of pressure or fullness in the epigastrium, abdominal pain and diarrhoea may occur in isolated cases, liver enzyme levels may increase, and impairment of liver function (e.g. with cholestasis and jaundice) and hepatitis may develop, possible resulting in liver failure.

Blood: Rarely thrombocytopenia and in isolated cases, leucopenia, hemolytic anaemia, erythrocytopenia, granulocytopenia, agranulocytosis, and pancytopenia (e.g. due to myelosuppression) may develop.

Other adverse reactions: Allergic or pseudoallergic reactions like itching, urticaria or rashes may occur. Such reactions are mild, but may become more serious and be accompanied by dyspnoea, and a fall in blood pressure, sometimes progressing to shock. If urticaria occurs, a physician must be notified immediately. In isolated cases allergic vasculitis, hypersensitivity of the skin to light, and a decrease in serum sodium may occur.

Metabolism and nutrition disorders

Common: Vitamin B12 decrease/deficiency

SYMPTOMS AND TREATMENT OF OVERDOSE:

Severe hypoglycaemic reactions, with coma, convulsions or other neurological disorders are possible and must be treated as medical emergency, requiring immediate hospitalization. If hypoglycaemic coma is diagnosed or suspected, the patient should be given a rapid IV injection of 50 ml of concentrated glucose solution (20 to 30 %). This must be followed by the infusion of more dilute glucose solution (10 %) at a rate that will maintain blood glucose levels above 1 g/L. Patients should be monitored closely and, depending on the patients condition after this time, the doctor will decide if further monitoring is necessary. Dialysis is not beneficial to patient due to strong binding of glimepiride to plasma proteins. Overdosage of sulphonylureas, including glimepiride, can produce hypoglycaemia. Mild hypoglycaemic symptoms without loss of consciousness or neurologic findings should be treated aggressively with oral glucose and adjustments in drug dosage and/or meal patterns. Close monitoring should continue until the physician is assured that the patient is out of danger.

Lactic acidosis is a rare, but serious, metabolic complication that can occur if metformin accumulates during treatment due to overdosing. Strict monitoring should be continued until the doctor is sure that the patient is out of danger.

Hemodialysis may be useful for removal of accumulated metformin from patients in whom overdosage is suspected.

Effects on Ability to Drive and Use Machine

No studies on the effects on the ability to drive and use machines have been performed.

The patient's ability to concentrate and react may be impaired as a result of hypoglycaemia or hyperglycaemia or, for example, as a result of visual impairment. This may constitute a risk in situations where these abilities are of special importance (e.g. driving a car or operating machinery).

Patients should be advised to take precautions to avoid hypoglycaemia whilst driving. This is particularly important in those who have reduced or absent awareness of the warning symptoms of hypoglycaemia or have frequent episodes of hypoglycaemia. It should be considered whether it is advisable to drive or operate machinery in these circumstances.

Instruction for Use

For Oral Use.

STORAGE CONDITIONS:

Store below 30°C. Protect from light and moisture. Keep out of reach of children.

SHELF LIFE:

24 months

PRESENTATION:

PROGLUTROL G2 Sustained Release Tablet 2/500 mg tablets are available as 20 tablets in a blister.

100 tablets in a box.

Manufactured by :

Inventia

Inventia Healthcare Limited,

F1-F1/1-F75/1, Additional Ambernath M.I.D.C., Ambernath (East),

Thane, Maharashtra State, Ambernath, 421506, India

Product registration holder in Malaysia:

Healol Pharmaceuticals Sdn. Bhd.

74-3, Jalan Wangsa Delima 6, KLSC Wangsa Maju, 53300 Kuala Lumpur, Malaysia

Date of Revision of Package Insert:

Apr. 2024

NOTE:

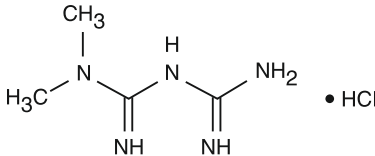
- Read the instructions thoroughly before use.
- Use upon doctor's prescription only.
- Please do not use the drug after the expiry date.
- Please do not use the drug if there are any significant changes in appearance of the tablets.
- Keep out of reach of children.

Package Insert

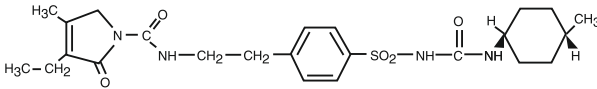
GLIMEPIRIDE 2MG AND METFORMIN
HYDROCHLORIDE SR 500MG TABLETS
PROGLUTROL G2
Sustained Release Tablet 2/500 mg

PRODUCT DESCRIPTION

PROGLUTROL G2 Sustained Release Tablet 2/500 mg is a brand of Glimepiride & Metformin HCl SR Tablets which is Capsule shaped biconvex, bilayered tablet, plain on both sides, with one layer white in colour & the other layer yellow and blister packed. Proglutrol G2 contains Metformin hydrochloride, an oral antihyperglycemic drug used in the management of type 2 diabetes. Metformin hydrochloride (N, N dimethylimidodicarbonimidic diamide hydrochloride) is a member of the biguanide class of oral antihyperglycemics and is not chemically or pharmacologically related to any other class of oral anti hyperglycemic agents. The empirical formula of metformin hydrochloride is C₄H₁₁N₅• HCl and its molecular weight is 165.63. Its structural formula is:



Glimepiride is a 1-[[p-[2-(3-ethyl-4-methyl-2-oxo-3-pyrrolidine-1-carboxamido) ethyl]phenyl]sulfonyl]-3-(trans-4-methylcyclohexyl)urea. It's a empirical formula is C₂₆H₃₈N₄O₄S, with a molecular weight of 490.62. The structural formula is:



COMPOSITION:

PROGLUTROL G2 Sustained Release Tablet 2/500 mg

Glimepiride 2 mg and Metformin Hydrochloride SR 500 mg Tablets

Each uncoated bilayered tablet contains:

Glimepiride 2 mg

Metformin Hydrochloride 500 mg

(in sustained release form)

Excipients q.s.

Colours: Ferric oxide (Yellow) and Titanium dioxide

Inactive Ingredients:

Hypromellose, Carboxymethylcellulose Sodium, Methacrylic acid copolymer dispersion, Macrogol, Povidone, Magnesium Stearate, Sodium starch glycolate, Lactose, Ferric oxide (yellow), Titanium dioxide, Microcrystalline cellulose, Pregelatinised Starch.

PHARMACODYNAMICS/PHARMACOKINETICS:

Pharmacotherapeutic group : Oral anti-diabetics

ATC code : i) Glimepiride

: A10BB12

ii) Metformin Hydrochloride

: A10BA02

Pharmacodynamics:

Glimepiride reduces blood glucose levels by correcting both defective insulin secretion and peripheral insulin resistance. It interacts with specific receptors at the plasma membrane of the insulin releasing pancreatic beta-cells where it inhibits ATP- sensitive K⁺ channels resulting in depolarization of the cell membrane, opening of voltage sensitive Ca²⁺ channels, increase in intracellular calcium levels and subsequent insulin release.

Metformin acts as an antihyperglycaemic agent by improving hepatic and peripheral tissue sensitivity to insulin. It also appears to have beneficial effect on serum lipid levels and so on fibrinolytic activity. Metformin therapy is not associated with increase in body weight. Metformin decreases glucose production, decreases intestinal absorption of glucose and improves insulin sensitivity by increasing peripheral glucose uptake and utilization.

Pharmacokinetics:

Glimepiride is rapidly and completely absorbed after oral administration. The oral bioavailability is approximately 100%. Following oral administration of 1 mg single dose to healthy volunteers, peak serum concentration (C_{max}) of 103 ± 34 ng/ml occurred within 2-3 hours. More than 99% of the drug is bound to plasma proteins. Glimepiride is completely biotransformed by hepatic oxidative metabolism into cydohexylhydroxymethyl derivative (M1) which is further metabolized to form a carboxyl derivative (M2) by cytosolic enzymes. After a single dose, the elimination half life (t_{1/2}) of glimepiride is 5 hours. The urinary excretion of metabolites accounts for 60% of dose, the remainder is found as metabolites in faeces.

Metformin has absolute oral bioavailability of 50-60%. GIT absorption is complete within 6 hrs of ingestion and Metformin is rapidly distributed in body after absorption. The renal elimination of Metformin is biphasic. 95% of the absorbed Metformin is eliminated during primary elimination phase having half-life of 6 hours. Rest of the 5% is eliminated during slow terminal elimination phase with mean half-life of 20 hours. Metformin is not bound to plasma proteins, 40-60% of the dose is recovered as unchanged drug in urine with a further 30% recovered

200 mm

200 mm

Product Name : PROGLUTROL G2 (Malaysia) (For Variation purpose)	
Packaging Material : Pack Insert (Front & Back Side)	
Location : Ambernath	Version No.:
Size : 200 x 200 mm (As per Drawing)	
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Artwork No.:	SAP Code:
Substrate : White rectangular piece of Maplitho paper	
Drawing No. : CL - 268 - R2	
Specification : 60 GSM	
Pharmacode : NA	
Any other Requirement : Unfolded and Transparency required	
Change History : 1) C/ - Text Added 2) Addition of Plot from "F1-F1/1 to F1-F1/1-F75/1" in Inventia address 3) Date of Revision of Package Insert changed from “May 2020” to “Apr. 2024”	

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as unchanged drug in faeces.

INDICATIONS AND USAGE:

Proglutrol G2 is indicated as an adjunct to diet and exercise in non-insulin dependent diabetes mellitus (NIDDM) (Type 2) patients who are unable to achieve sufficient glycaemic control with monotherapy of metformin or glimepiride alone or who are already treated with combination of metformin and glimepiride as separate tablets.

RECOMMENDED DOSAGE AND ADMINISTRATION:

Glimepiride 2 mg/ Metformin hydrochloride SR 500 mg: 1 tablet once daily or as directed by the physician. Dosage must be individualized on the basis of both effectiveness and tolerance, while not exceeding the maximum recommended daily dose. The highest recommended dose per day should be 8 mg of glimepiride and 2000 mg of metformin. Daily doses of glimepiride of more than 6 mg are more effective only in a minority of patients. Do not crush or chew the tablet; the whole tablet to be taken with water. Start with one tablet per day. The aim should be to decrease both fasting plasma glucose and glycosylated hemoglobin levels to normal by using the lowest effective dose of the drug.

Renal Impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression or renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months. Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin in patients with GFR<60 mL/min. If no adequate strength of Proglutrol G2 Sustained release Tablet 2/500mg is available, individual monocomponents should be used instead of the fixed dose combination.

GFR mL/min	Metformin	Glimepiride
60-89	Maximum daily dose is 2000mg. Dose reduction may be considered in relation to declining renal function	Maximum daily dose is 8 mg. Daily doses of more than 6 mg are more effective only in a minority of patients. Glimepiride is contraindicated in patients with severe renal function disorders.
45-59	Maximum daily dose is 2000mg. The starting dose is at most half of the maximum dose.	
30-44	Maximum daily dose is 1000mg. The starting dose is at most half of the maximum dose.	
<30	Metformin is contraindicated.	

MODE OF ADMINISTRATION:

The mode of administration is oral.

CONTRA-INDICATIONS:

This combination is not suitable for the treatment of insulin-dependent (type 1) diabetes mellitus (e.g. For the treatment of diabetics with a history of ketoacidosis), or of diabetic precoma or coma. It must not be used in patients hypersensitive to metformin hydrochloride, glimepiride, sulfonylureas, other sulfonamides, or any of the excipients (risk of hypersensitivity reactions). Impaired renal function Acute complications (severe infections, major operations, and trauma) Before X-ray examinations with iodinated contrast materials. Liver damage Alcoholism Deficiencies of vitamin B12, folic acid, and iron Ketosis-prone diabetes Severe cardiovascular or respiratory disease General ill health (malnutrition, dehydration, etc) Diabetes with significant late complications (nephropathy, retinopathy) Severely reduced kidney function (GFR <30 mL/min) Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)

WARNINGS AND PRECAUTIONS:

Glimepiride

If risk factors for hypoglycemia are present, it may be necessary to adjust the dosage of glimepiride or the entire therapy. This also applies whenever illness occurs during therapy or the patient's life-style changes. Symptoms of hypoglycemia may be milder or absent in those situations where hypoglycemia develops gradually, in the elderly, and in the patients with autonomic neuropathy or those receiving concurrent treatment with beta-blockers, clonidine, reserpine, guanethidine or other sympatholytic drugs. Hypoglycemia can almost always be promptly controlled by immediate intake of carbohydrates (glucose or sugar, e.g. in the form of sugar lumps, sugar-sweetened fruit juice or sugar sweetened tea). For this purpose, patients must carry a minimum of 20 grams of glucose with them at all times. They may require assistance of other persons to avoid complication. Artificial sweeteners are ineffective in controlling hypoglycemia. Continued dose observation is necessary. Severe hypoglycemia requires immediate treatment and follow - up by a physician and in some circumstances, hospitalization. In exceptional stress situations (e.g. trauma, surgery, infections with fever) blood sugar control may deteriorate, and a temporary change to insulin may be necessary. During treatment with glimepiride, glucose levels in blood and urine must be checked regularly, as should, additionally, the proportion of glycated hemoglobin. Alertness and reactions may be impaired due to hypo- or hyperglycemia, especially when beginning or after altering treatment, or when glimepiride is not taken regularly. This may affect the ability to operate vehicle or machinery.

Metformin

Lactic Acidosis: Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal

function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5mmol/L) and an increased anion gap and lactate/ pyruvate ratio.

Metformin can provoke lactic acidosis; however, the reported incidence is very low. Conditions like impaired hepatic function, renal dysfunction, hypoxemia, dehydration, sepsis, excessive alcohol intake can increase the risk of lactic acidosis. The risk can be decreased by regular monitoring of renal function and by use of minimum effective dose. In a patient with lactic acidosis, who is on metformin treatment, the drug should be discontinued immediately. Supportive measures & prompt hemodialysis to be started. Impaired renal function: Caution should be exercised with concomitant therapies that may affect renal function or interfere with the disposition of metformin (e.g. cationic drugs).

Metformin may reduce vitamin B 12 serum levels. The risk of low vitamin B 12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B 12 monitoring could be necessary in patients with risk factors for vitamin B 12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines

Renal function

GFR should be assessed before treatment initiation and regularly there after [See Section Recommended Dosage]. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that after renal function [See Section Contraindications] Use of iodinated contrast media: The drug should be stopped at least two days before X-ray examination with iodinated contrast material and reinstituted only after renal function has been re-evaluated and found to be normal.

Hypoxic states: Metformin therapy should be promptly discontinued when such events occur in patients.

Surgical procedures:

The drug should be temporarily discontinued and restarted only when the patient resumes oral intake and has normal renal function.

Alcohol intake: Patients to be warned against excessive alcohol intake, acute or chronic, while receiving metformin.

Impaired hepatic function: The drug should be generally avoided in patients with hepatic disease.

Hypoglycemia:

Does not normally occur when the drug is given alone but has been observed when given in combination with sulfonylureas and /or alcohol.

Deficiencies of folic acid, iron and Vitamin B12: Serum Vitamin B12 concentrations should be measured annually during long-term treatment.

Laboratory Tests:

Monitoring of response to therapy to be done periodically through measurement of fasting blood glucose and glycosylated hemoglobin levels. During initial dose titration, fasting glucose can be used to determine the response. Subsequently, both glucose and glycosylated hemoglobin must be monitored, which may be useful in evaluating long-term control.

DRUG INTERACTIONS:

Cimetidine:

Metformin interacts with cimetidine. Therefore, the dose of metformin should be reduced if cimetidine is co-prescribed.

Hyperglycemic agents:

Drugs with hyperglycemic potential (e.g. thiazides, corticosteroids, and others) may partly offset the anti-hyperglycemic action of metformin, and in such cases the glycemic control should be closely monitored.

Alcohol:

Alcohol potentiates the action of metformin on lactate metabolism as well as the anti-hyperglycemic effect. Hence, patients treated with metformin should preferably avoid alcohol, and alcoholism is a definite contraindication.

Studies with furosemide and nifedipine suggest a possible interaction by increasing plasma metformin levels. However no such changes were found with propranolol and ibuprofen. The absorption of metformin may be reduced by acarbose and guar gum. Hypoglycemia due to interaction with glimepiride may occur when one of the following medicines is taken, for example: insulin and other oral antidiabetics, ACE inhibitors, allopurinol, anabolic steroids and male sex hormones, chloramphenicol, coumarin derivatives, cyclophosphamide, disopyramide, fenfluramine, fenyramidol, fibrates, fluoxetine, guanethidine, ifosfamide, MAO inhibitors, miconazole, para-aminosalicylic acid, pentoxifylline (high dose parenteral), phenylbutazone, azapropazone, oxypentenbutazone, probenecid, quinolones, salicylates, sulfapyrazone, sulfonamides, tetracyclines, tritoqualine, trofosamide. Hyperglycemia due to interaction with glimepiride may occur when one of the following medicines is taken, for example: acetazolamide, barbiturates, corticosteroids, diazoxide, diuretics, epinephrine (adrenaline) and other sympathomimetic agents, glucagons, laxatives (after protracted use), nicotinic acid (in high doses), oestrogens and progestogens, phenothiazines, phenytoin, rifampicin, thyroid hormones. H2 receptor antagonists, clonidine and reserpine may lead to either potentiation or weakening of the blood sugar lowering effect. Beta-blockers may increase the tendency to hypoglycemia. The effect of coumarin derivatives may be potentiated or weakened.

PREGNANCY AND LACTATION:

Pregnancy is generally regarded as a contra-indication and insulin should be used in all pregnant diabetic women. Do not take this medicine if you are pregnant or plan to become pregnant.